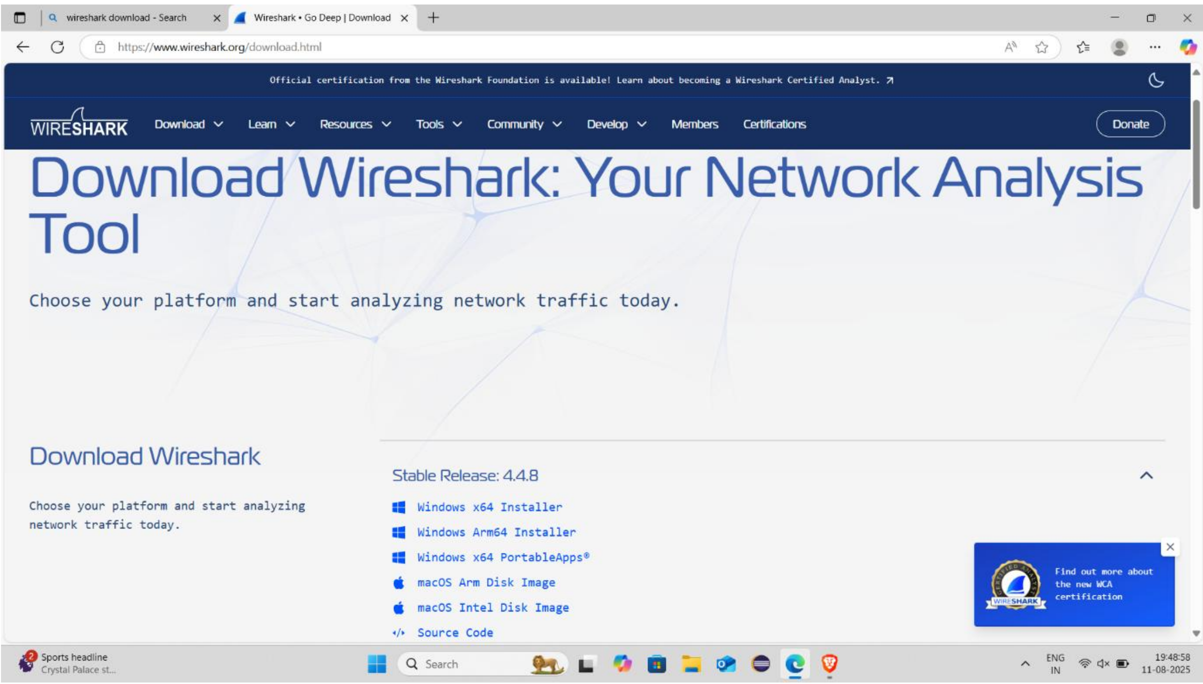
Task 5 : Capture and Analyze Network Traffic Using Wireshark.

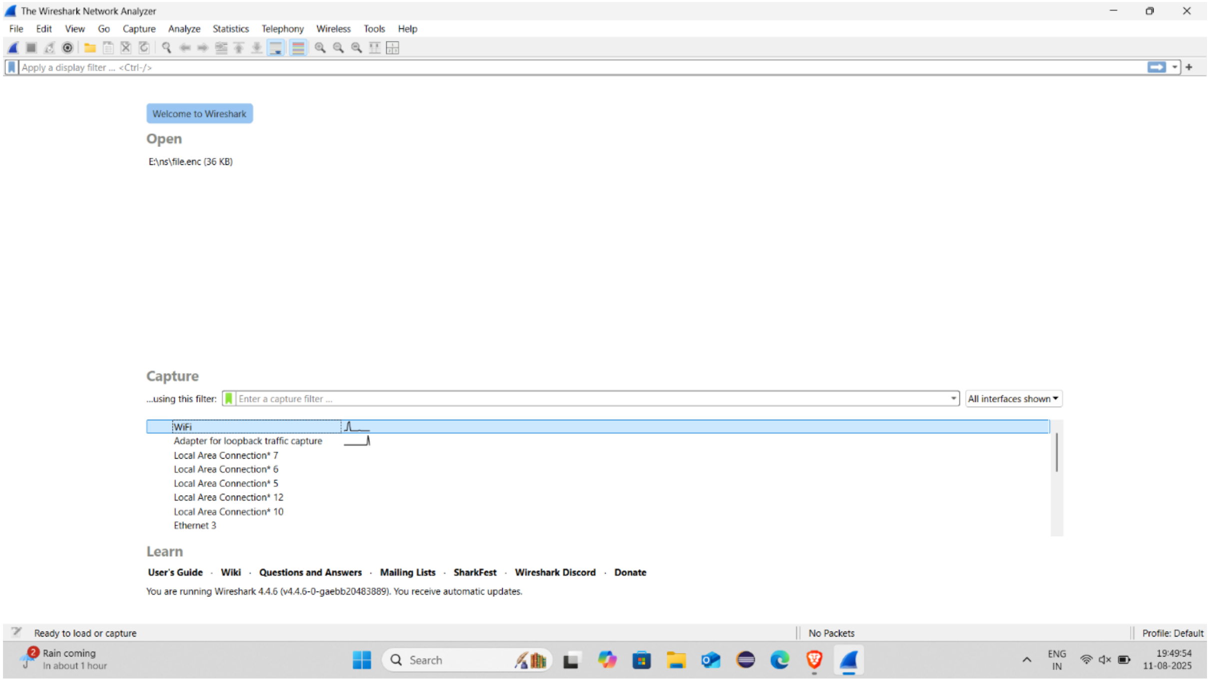
Step 1 – Install Wireshark

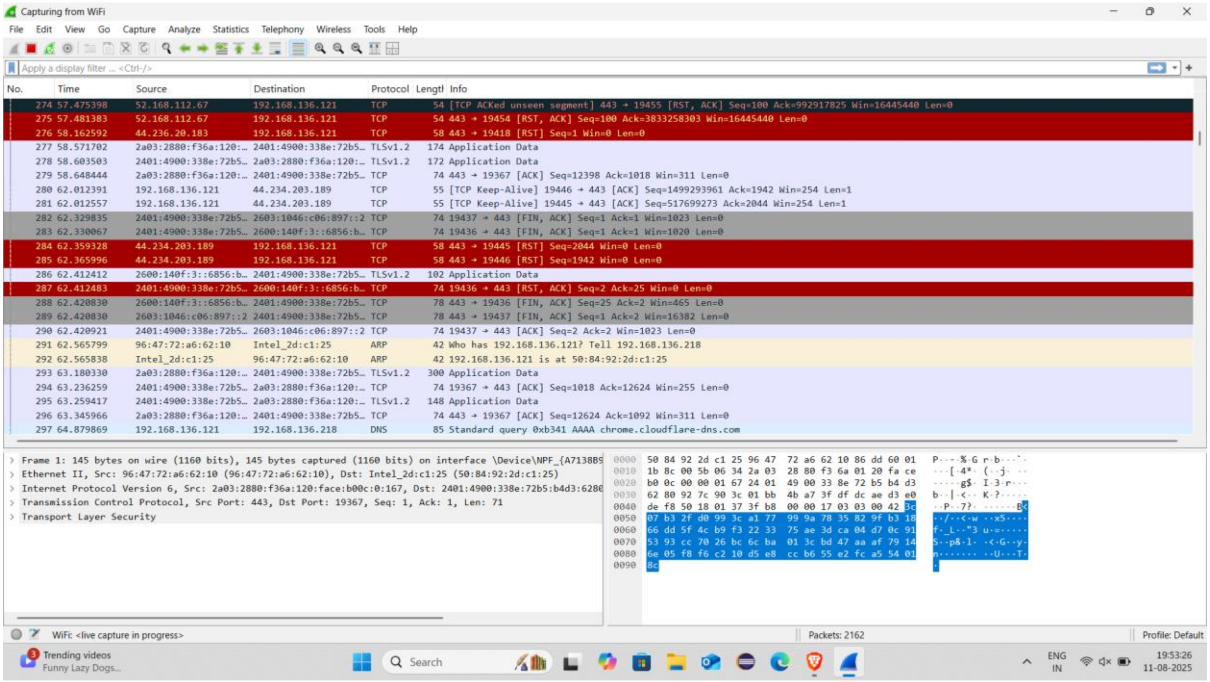
1. Go to the Wireshark website.
2. Download the installer for your OS (Windows, Linux, macOS).
3. Install Wireshark with default settings.
   - On Windows, allow installation of **Npcap** (needed for packet capture).



Step 2 – Start a Packet Capture

1. Open Wireshark.
2. You'll see a list of network interfaces (Ethernet, Wi-Fi, etc.).
3. Select your **active network interface** (the one showing live traffic).
4. Click the **blue shark fin** button (Start Capturing).





Step 3 – Generate Network Traffic

To make your capture more interesting:

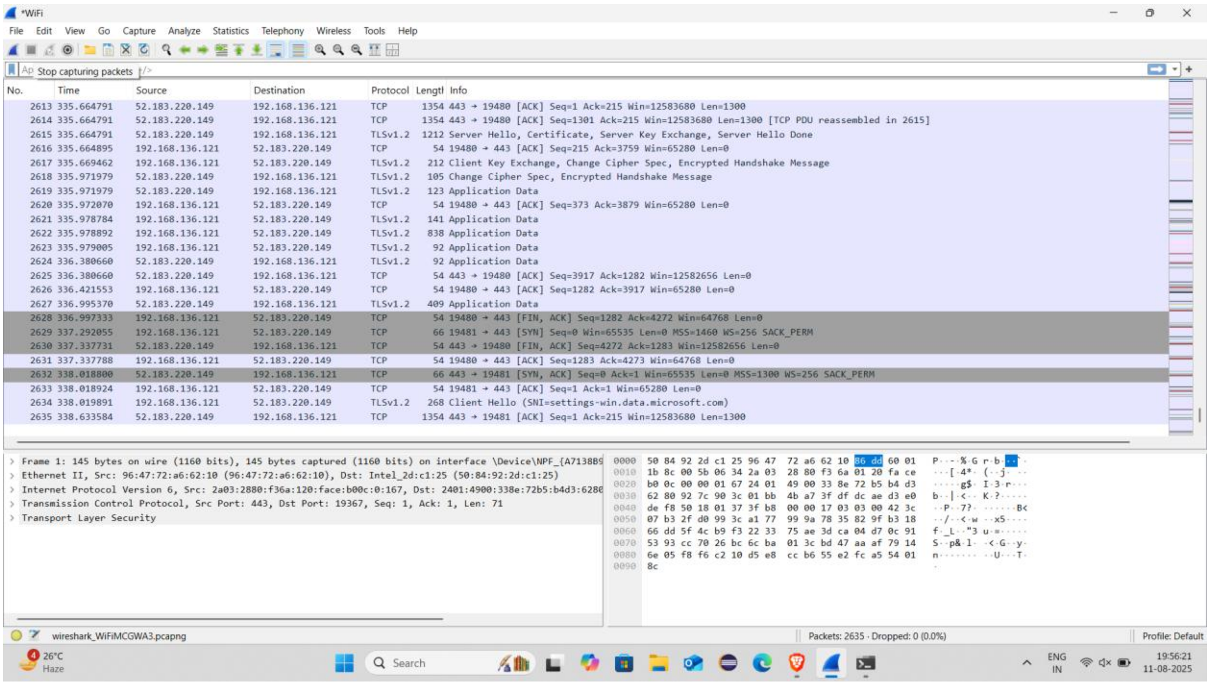
- Open a browser and visit a few websites.
- You can also send an email, watch a short video, etc.

- Or, in Command Prompt/Terminal, run:

```
C:\WINDOWS\system32\cmd.

Microsoft Windows [Version 10.0.26100.4351]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hari6>ping www.google.com

Pinging www.google.com [2404:6800:4007:82e::2004] with 32 bytes of data:
Reply from 2404:6800:4007:82e::2004: time=80ms
Reply from 2404:6800:4007:82e::2004: time=124ms
Reply from 2404:6800:4007:82e::2004: time=93ms
Request timed out.

Ping statistics for 2404:6800:4007:82e::2004:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 80ms, Maximum = 124ms, Average = 99ms

C:\Users\hari6>
```

Step 4 – Stop Capture

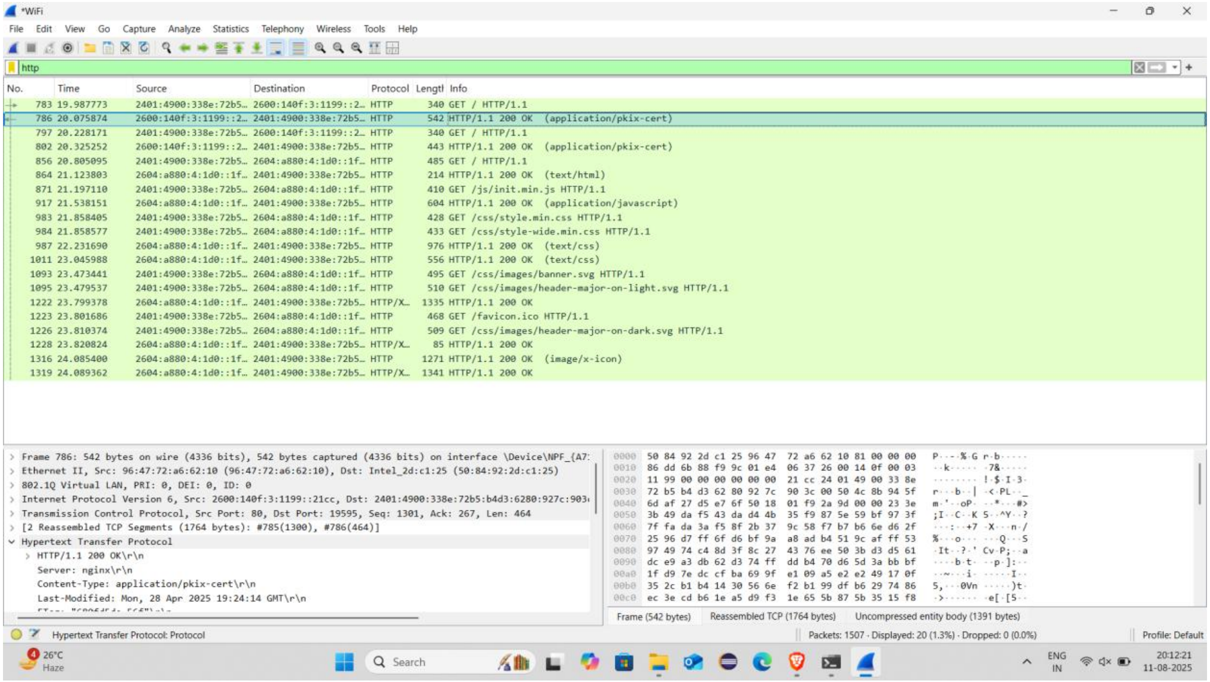
1. Let it run for **about 1 minute**.
2. Click the **red square (Stop)** button



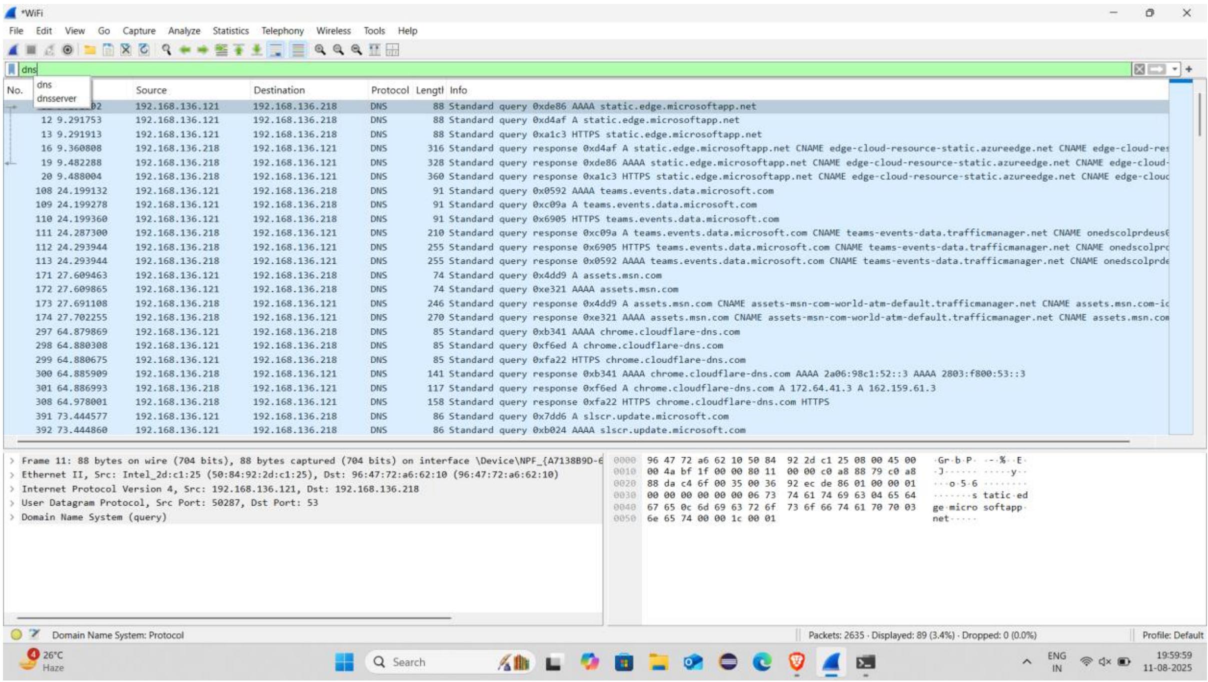
Step 5 – Filter by Protocol

Wireshark has a filter bar at the top. Some useful examples:

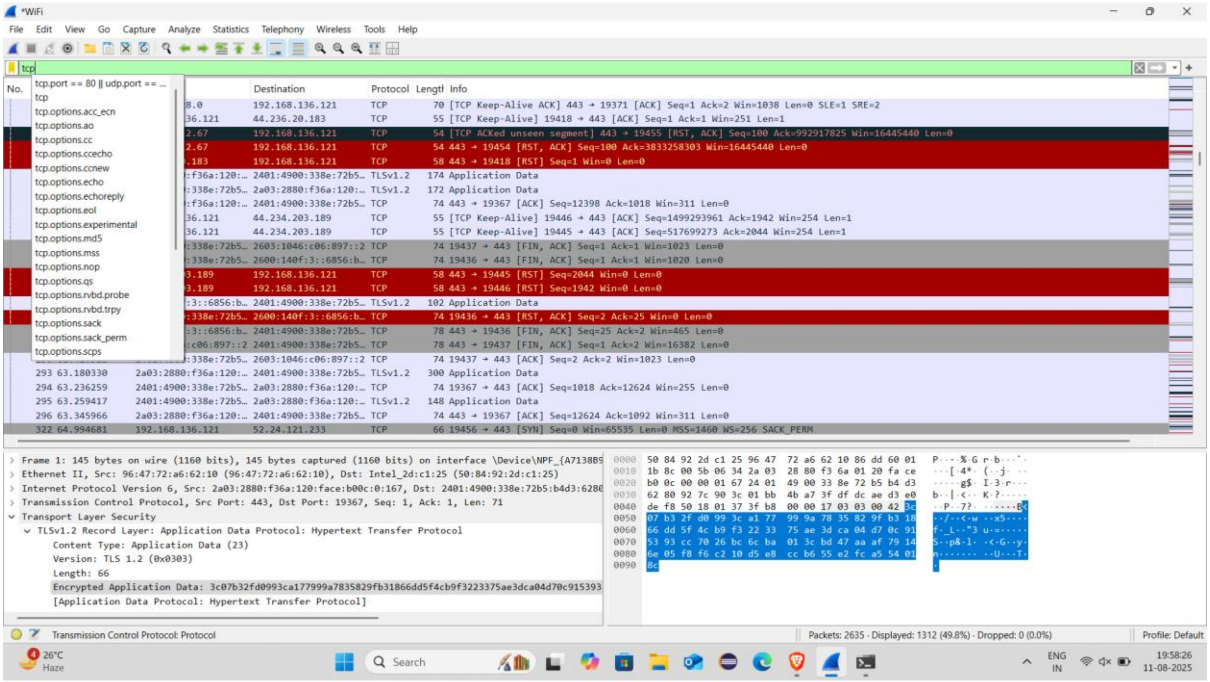
- **HTTP traffic:**



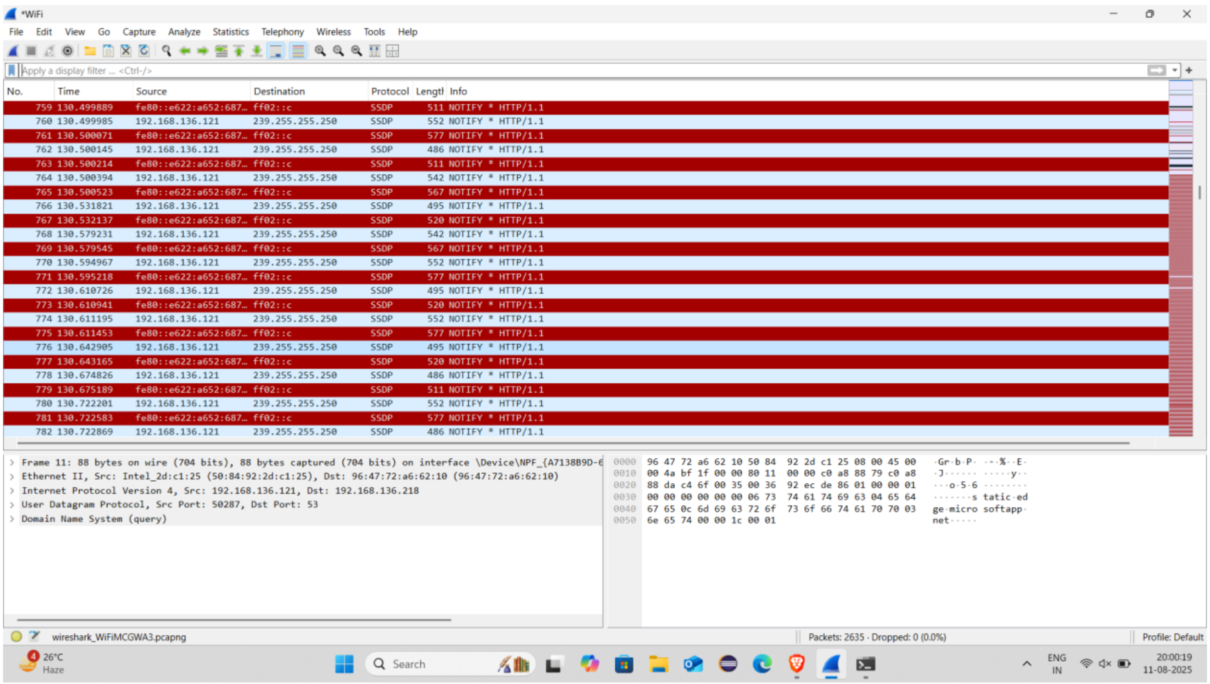
- **DNS traffic**

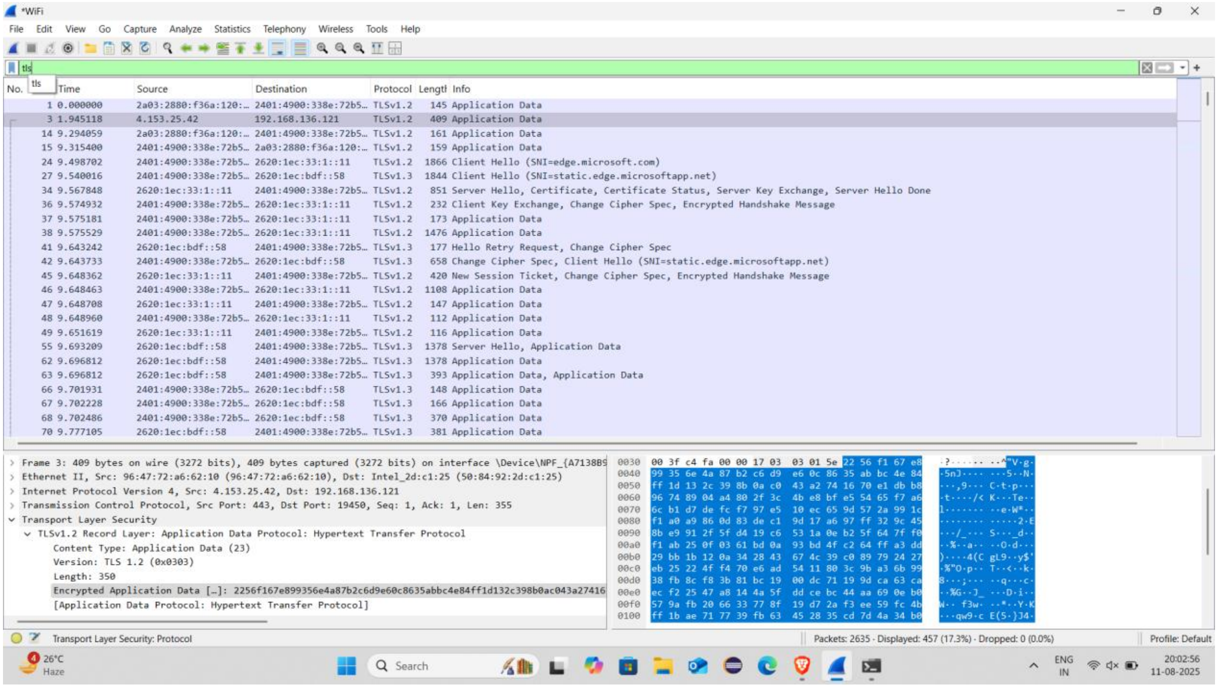


- **TCP traffic:**



- **Others(TLS & SSDP)**

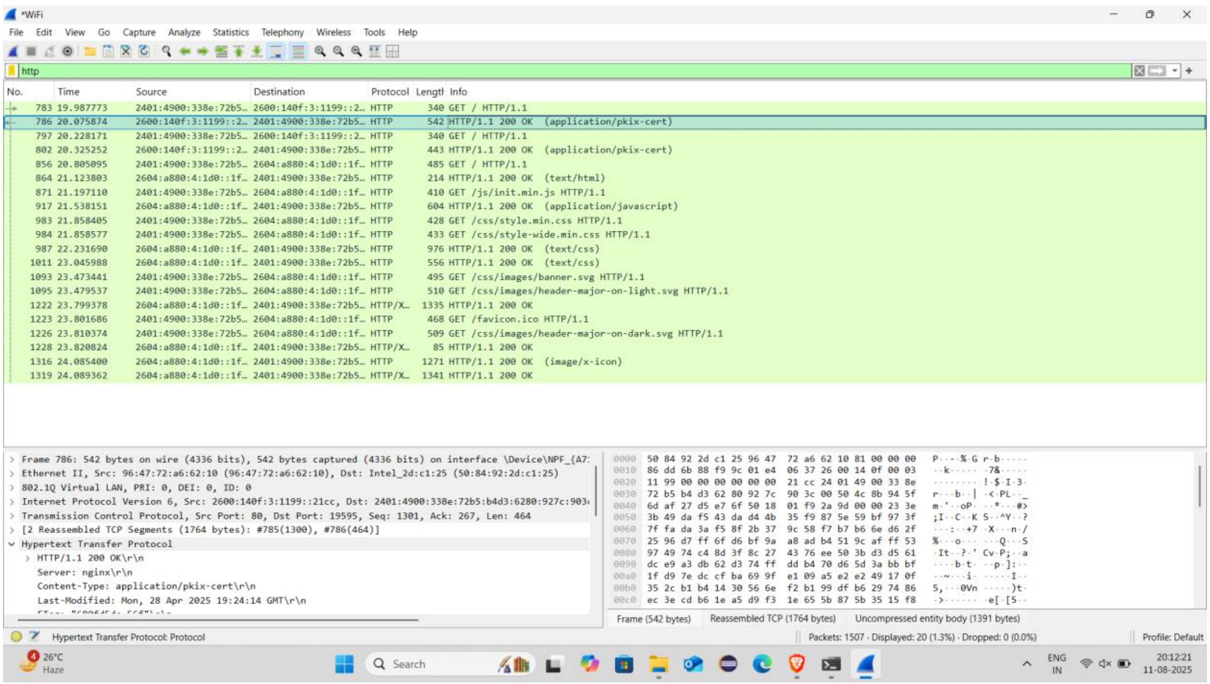




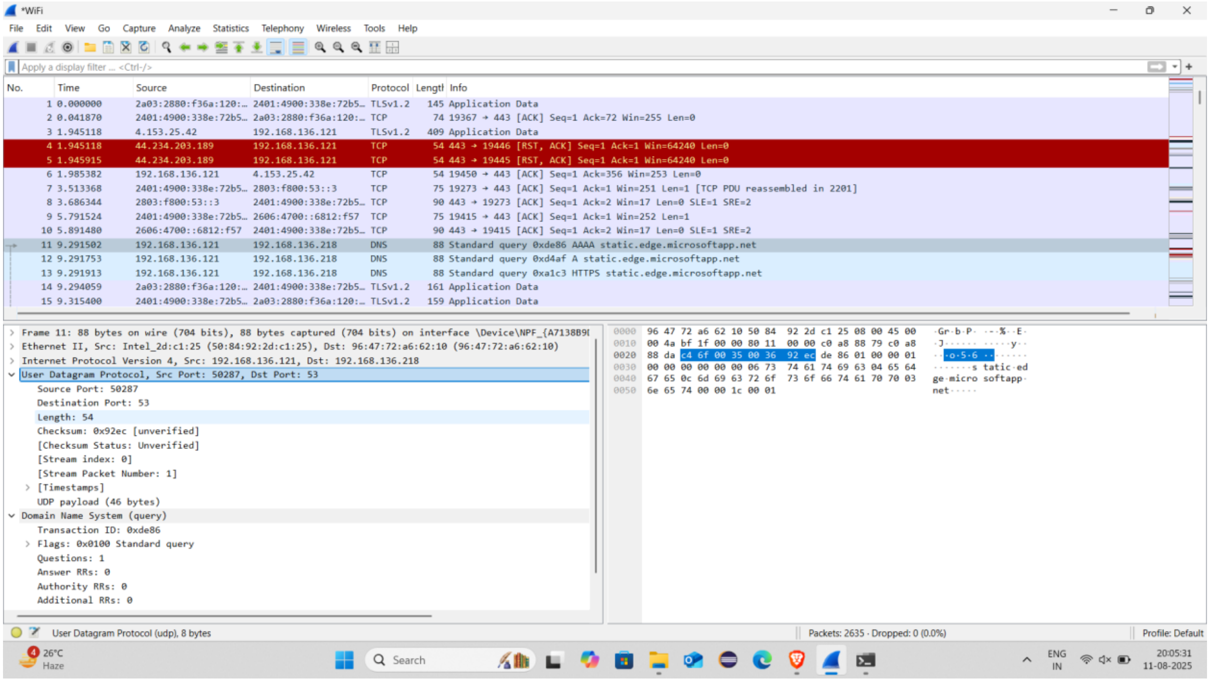
Step 6 – Identify at Least 3 Protocols

Look for:

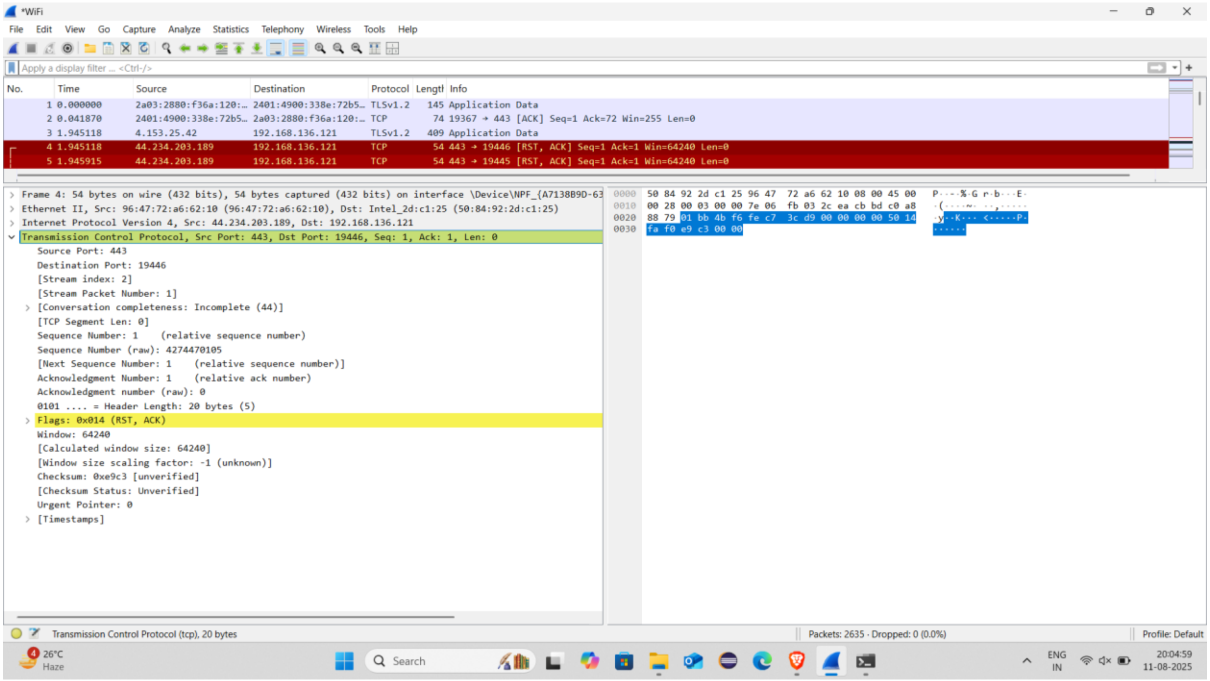
- **HTTP** → web browsing data.



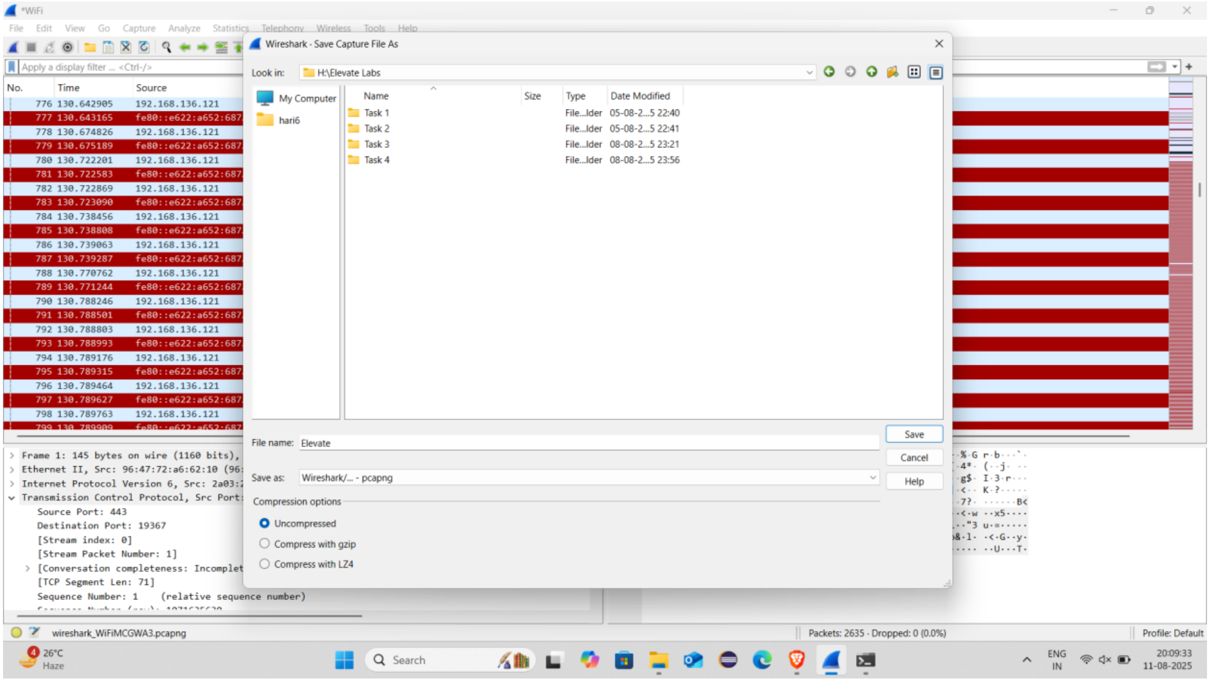
- **DNS** → domain name lookups.



- **TCP**→ transport layer communication.



Step 7 – Export the Capture as `.pcap`



1. Go to **File → Save As**.
2. Choose `.pcapng` or `.pcap` format.
3. Name your file and save it for analysis or sharing.